Anna Chrzanowska

Paris, France | anna.chrzanowska@icm-institute.org | aniachrzanowska.github.io
anna-chrzanowska-b6101b30a | Google Scholar

Summary

Experienced neuroscientist with 10+ years of expertise in behavioral systems neuroscience. Specialized in a range of rodent behavioral assays, neural circuit genetic manipulations, and neural activity recordings (both *in vivo* freely-moving and head-fixed electrophysiology). Skilled in experimental setup design, optimization, and troubleshooting, as well as performing microsurgeries. Demonstrated success in securing fellowships, admissions to competitive training programs, and travel grants. Proven leadership in mentoring Master's and Bachelor's students. Actively involved in organizing symposia and conferences, and in science popularization. Engaged member of the international community of young neuroscientists.

Skills

Technical: Rodent behavioral assays, neural circuit genetic manipulations, *in vivo* electrophysiology (freely-moving & head-fixed), two-photon calcium imaging, microsurgeries, brain histology, viral injections, optogenetics, chemogenetics, patch clamp recordings (*in vitro*).

Engineering & Software: Experimental setup design, optimization, troubleshooting, Bonsai reactive programming, Arduino, electronics, virtual reality environments for mice.

Professional: Mentoring, project supervision, scientific writing, manuscript editing, organizing symposia and conferences, science popularization.

Education

PhD **Arenberg Doctoral School, KU Leuven**, Biology — Leuven, Belgium, Jan 2018 – Aug 2023

- Dissertation: Unraveling the cell-type specific underpinnings that shape collicular-mediated visual behaviors
- Supervisor: Dr. Karl Farrow (Neuro-Electronics Research Flanders, NERF)

MSc **Jagiellonian University**, Neurobiology — Cracow, Poland, Oct 2015 – Nov 2017

- Dissertation: Circadian analysis of the orexin fibers distribution in the rat lateral geniculate nucleus of the thalamus
- Supervisor: Prof. Dr. Marian H. Lewandowski

BSc **Jagiellonian University**, Neurobiology — Cracow, Poland, Oct 2012 – July 2015

- Dissertation: Role of GABA-ergic innervation in the mammalian biological clock
- Supervisor: Prof. Dr. Marian H. Lewandowski

Research Experience

Bassem Hassan Lab, ICM – Paris Brain Institute, Postdoctoral Researcher — Paris, France, Jan 2024 – present

Investigating links between brain wiring structure, physiology, and their significance for behavior.

- Neurodevelopmental fly genetics; behavioral experiments
- *In vivo* two-photon calcium imaging in behaving flies

Marion Silies Lab, Johannes Gutenberg University of Mainz, Visiting Scientist — Mainz, Germany, Mar 2025

- Data analysis of calcium imaging signals

Karl Farrow Lab, NERF (imec, KU Leuven & VIB), PhD Researcher — Leuven, Belgium
Investigating the role of distinct neuronal cell types in visually-guided innate behaviors. — Jan 2018 – Aug 2023

- Open-field behavioral paradigms to study innate behaviors in mice
- *In vivo* freely-moving and head-fixed Neuropixels recordings
- Cell-type specific optogenetics and chemogenetics; viral injections

Jagiellonian University (Marian H. Lewandowski Lab), Undergraduate Researcher — Cracow, Poland
Investigating circadian changes of orexinergic innervation in the rat visual system. — Oct 2012 – Nov 2017

- Brain histology; assistance with *in vitro* patch clamp recordings

Karl Farrow Lab, NERF, Research Intern — Leuven, Belgium
Establishing a behavioral protocol for visually-triggered orienting and defensive behaviors in virtual reality. — Mar 2017 – July 2017

- Virtual reality environment for mice; head-fixed behavioral protocols

Behavioral Neurodynamics Lab (Ponomarenko / Korotkova), Leibniz-Institute for Molecular Pharmacology (FMP), Research Intern — Berlin, Germany, Oct 2016 – Feb 2017
Investigating sleep and memory regulation by forebrain circuits.

- Projection- and cell-type specific optogenetic interrogation
- Local field potential recordings *in vivo* freely-moving behavioral protocols

Laboratory of Neurodegeneration (Kuźnicki Lab), International Institute of Molecular and Cell Biology, Research Intern — Warsaw, Poland, Sept 2016

- Development of *in situ* proximity ligation assay (PLA) on brain slices for Huntington's disease research

Laboratory of Cell Biology of the Synapse (Matteoli Lab), Neuroscience Institute, CNR, Erasmus+ Trainee — Milan, Italy, Jan 2015 – Mar 2015

- Investigating the role of Eps8, an actin regulatory protein, in synapse plasticity; Western blot analysis

Awards and Distinctions

- Carnot Training Grant for short collaborative visits (2025)
- Marie Skłodowska-Curie Actions Postdoctoral Fellowship (2025)
- French Foundation for Medical Research (FRM) Postdoctoral Fellowship (2024)
- Fyssen Foundation Postdoctoral Fellowship (2023)
- Transylvanian Experimental Neuroscience Summer School (TENSS), participant (2021)
- CAJAL NeuroKit Course, participant (2021)
- Research Foundation – Flanders (FWO) PhD Fellowship, ~20% success rate (2018–2022)
- Travel Grant — CAJAL Advanced Neuroscience Training Programme (2018)
- Travel Grant — 7th Neuron Technology Summer School (2017)
- Deans' Award for Outstanding Academic Achievement (2015/16, 2016/17)
- Laureate — "Grasz o staz" (Win the Internship) competition (2016)
- Best Theoretical Poster — 5th Aspects of Neuroscience (2015)
- Best Theoretical Poster — 4th Aspects of Neuroscience (2014)

Publications

Rethinking naturalistic behavior: The Terzolas Manifesto — 2025
Anna Chrzanowska, Mateusz Kostecki, Natalia Krasilshchikova, Matilde Perrino, Luigi Petrucco
10.31219/osf.io/ch3an_v2 (The Open Science Framework)

Multiplexed surface electrode arrays based on metal oxide thin-film electronics for high-resolution cortical mapping 2023

Horacio Londoño-Ramírez, Xiaohua Huang, Jordi Cools, Anna Chrzanowska, Clément Brunner, Marco Ballini, Luis Hoffman, Soeren Steudel, Cédric Rolin, Carolina Mora Lopez, Jan Genoe, Sebastian Haesler

10.1002/advs.202308507 (Advanced Science)

Optogenetic fUSI for brain-wide mapping of neural activity mediating collicular-dependent behaviors 2021

Arnau Sans-Dublanc, *Anna Chrzanowska*, Katja Reinhard, Dani Lemmon, Bram Nuttin, Théo Lambert, Gabriel Montaldo, Alan Urban, Karl Farrow

10.1016/j.neuron.2021.04.008 (Neuron (*equal contribution; co-lead author))

Research culture: science from bench to society 2021

Lorenzo Canti, Anna Chrzanowska, M. Giulia Doglio, Lia Martina, Tim Van Den Bossche

10.1242/bio.058919 (Biology Open (all authors contributed equally))

Multiple excitatory actions of orexins upon thalamo-cortical neurons in dorsal lateral geniculate nucleus 2017

Lukasz Chrobok, Katarzyna Palus, Anna Chrzanowska, Mariusz Kępczyński, Marian Lewandowski

10.1038/s41598-017-08202-8 (Scientific Reports)

Disinhibition of the intergeniculate leaflet network in the WAG/Rij rat model of absence epilepsy 2017

Lukasz Chrobok, Katarzyna Palus, Jagoda S. Jeczmien-Lazur, Anna Chrzanowska, Mariusz Kępczyński, Marian Lewandowski

10.1016/j.expneurol.2016.12.014 (Experimental Neurology)

The role of collicular cell types in behavioral reactions to different threat intensities in mice 2025

Bram Nuttin, *Anna Chrzanowska*, Amber Bruynsteen, Arnau Sans-Dublanc, Karl Farrow

(In preparation (*equal contribution; co-lead author))

Presentations

- **Invited Speaker** — How to write a successful MSCA proposal. PORT for Health Neuroscience, Wroclaw, Poland (2025)
- **Invited Speaker** — Cell vibes: how collicular cell-types contribute to guiding mice innate behaviors. Neurons in Action, Warsaw, Poland (2025)
- **Oral** — Towards embracing brain complexity: lessons from cell types of the superior colliculus. ENCODS, Paris, France (2022)
- **Oral** — A cell-type specific understanding of innate defensive behaviors: the collicular perspective. Neuromatch 3.0 (2020)
- **Poster** — Neural circuits of individuality in the context of learning. VIP & DIF, Paris, France (2024)
- **Poster** — Locomotion shapes visually-evoked neural dynamics through modulation of collicular populations. COSYNE, Lisbon, Portugal (2024)

Student Supervision

Master's Students: Helin Polat, Hala Rachel El Hajj (2024–2025); Chaimae Ouriaghli Taouil (2021–2022); Liuyin Yang, Nidaa Elsayed (2019–2020); Carolina Tigre Avelar (2017–2018)

Bachelor's Students: Sofia Rebiha Decesaro (2024); Diana Dayekh (2023–2024); Martyna Palys, Sade Adeyemi (2022–2023); Diethe Peeters, Tom Voet, Mathis Van den Bruel (2021–2022); Mathilda Corcoles (2020–2021)

Major Collaborations

- **Alan Urban, NERF** — established novel technology to study cell-type specific brain-wide circuits in mice (opto-fUSI).
- **Nanoengineering team, imec** — tested *in vivo* a novel device for large-scale brain recordings.

Scientific Community

- Teaching assistant, Transatlantic Behavioral Neuroscience Summer School (2021 – present)
- Local teaching assistant, CAJAL NeuroKit Course at NERF (2022)